

Construction of Equivalence

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1 Introduction (ND)

Fix G as a finite group. Let P_G be the *associativity* G -operad in $\mathbf{Set}_G$, defined by $P_G(n) := \mathbf{Set}(G, \Sigma_n)$, with structure maps defined by post composition

$$\gamma : \mathbf{Set}(G, \Sigma_k) \times \prod_{r=1}^k \mathbf{Set}(G, \Sigma_{j_r}) \cong \mathbf{Set} \left(G, \Sigma_k \times \prod_{r=1}^k \Sigma_{j_r} \right) \xrightarrow{\gamma \circ -} \mathbf{Set}(G, \Sigma_j),$$

, where $j = \sum_{r=1}^k j_r$, and $\gamma : \Sigma_k \times \prod_{r=1}^k \Sigma_{j_r} \rightarrow \Sigma_j$ is the structure map for associativity operad $\mathcal{M}$ in $\mathbf{Set}$. This is naturally a free $(G \times \Sigma)$ -operad, as the right $(G \times \Sigma_n)$ -action at each $P_G(n)$ is free.

However, it is mentioned in [1] that such G -operad is not finitely generated, as a rough explanation, for every $n \in \mathbb{N}$, there exists $m > n$ and a function $f : G \rightarrow \Sigma_m$, such that it cannot be generated using $\{P_G(k)\}_{k \leq n}$ with the operadic structure map and the $(G \times \Sigma)$ -actions.

As a substitution, [1] defines a finitely-generated suboperad $Q_{G, \mathfrak{D}} \hookrightarrow P_G$ (for some parameter $\mathfrak{D}$ that's associated with choice of transitive G -sets), such that after chaotic generalization the two are equivalent in $\mathbf{Cat}_G$, and having equivalent categories of operadic algebras.

We'll mimic the statement, define an intermediate operad $Q_{G, \mathfrak{D}} \hookrightarrow Q_G \hookrightarrow P_G$, promote it to $\mathcal{Q}_G$ in $\mathbf{Cat}_G$, and *attempt to* modify the proof in [2] to show the equivalence of 2-categories between $\mathcal{Q}_G\text{-PsAlg}$ and $\mathcal{F}_{G, \mathbb{R}}\text{-PsAlg}$.

Future task (assume this works): Prove the equivalence of operads/algebras/pseudoalgebras with the original permutativity G -operad.

Current Goal: Add in extra axioms, so that there exist two 2-functors

$$\mathbb{R}_G : \mathcal{Q}_G\text{-PsAlg} \rightleftarrows \mathcal{F}_{G, \mathbb{R}, -}\text{-PsAlg} : \mathbb{Q}_G$$

inducing an isomorphism of (strict) 2-categories. Here, $-$ is some (yet to be determined) axiom.

2 Preliminaries

2.1 Symmetric Monoidal Categories

2.2 Operads in Symmetric Monoidal Categories

2.3 Category $\mathcal{F}_G$

2.4 Equivariance Preliminaries

2.5 G -Categories

2.6 2-Categories

3 Essential Categorical Tools

Here we introduce some definitions important to this paper.

3.1 G -Operad $\mathcal{Q}_G$ in $\mathbf{Cat}_G$

Definition 3.1. Let Q_G be the suboperad of P_G , generated by all the group homomorphisms $\alpha : G \rightarrow \Sigma_n$ spanning through all $n \in \mathbb{N}$, the P_G structure maps γ_G , and the $(G \times \Sigma)$ -action on the elements. Define the operad $\mathcal{Q}_G := \mathcal{E}Q_G$ as the chaotic generalization in $\mathbf{Cat}_G$.

This operad is designed as a smallest suboperad that contains all group homomorphisms from G to a symmetric group, such that it's still a free $(G \times \Sigma)$ -operad.

Potential (fun) question: Is this suboperad finitely generated, or equals to any $\mathcal{Q}_{G,\mathfrak{D}}$ described in [1]?

Warning 3.2. If naively taken all group homomorphisms and their orbits under $(G \times \Sigma)$ -action, it doesn't form a suboperad in general for nontrivial group G . A counterexample can be constructed even for $G = \mathbb{Z}/2\mathbb{Z}$.

3.2 The 2-Category $\mathcal{Q}_G\text{-PsAlg}$

We mostly follow [2] in defining $\mathcal{Q}_G$ -pseudoalgebras.

Definition 3.3 ($\mathcal{Q}_G$ -pseudoalgebra). A $\mathcal{Q}_G$ -pseudoalgebra is a G -category $\mathcal{A}$ and operad action functors

$$\theta_j : \mathcal{Q}_G(j) \times \mathcal{A}^j \rightarrow \mathcal{A}$$

for each $n \in \mathbb{N}$, together with an invertible natural transformation ϕ_1 in the diagram

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{A} \\ & \searrow \phi_1 & \downarrow \theta_1 \\ & \text{Id}_{\mathcal{A}} & \mathcal{C} \end{array} \quad (3.4)$$

and a natural transformation $\phi = \phi(k; j_1, \dots, j_k)$ for each diagram

$$\begin{array}{ccc} \mathcal{Q}_G(k) \times \left(\prod_{r=1}^k \mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r} \right) & \xrightarrow{\text{id} \times \prod_r \theta_{j_r}} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\ \downarrow \text{shuffle} & \searrow \phi & \downarrow \theta_k \\ \mathcal{Q}_G(k) \times \prod_{r=1}^k \mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r} & \xrightarrow{\gamma \times \text{id}} & \mathcal{Q}_G(j_+) \times \mathcal{A}^{j_+} \end{array} \quad (3.5)$$

such that the following strict equivariance hold for θ_j and ϕ :

- (i) **(Equivariance of θ_j)** The following diagram commutes for $(g, \sigma) \in G \times \Sigma_j$:

$$\begin{array}{ccc} \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{(g, \sigma) \times (g, \sigma)^{-1}} & \mathcal{Q}_G(j) \times \mathcal{A}^j \\ & \searrow \theta_j & \swarrow \theta_j \\ & \mathcal{A} & \end{array} \quad (3.6)$$

(ii) **(Equivariance of ϕ)** The following equalities hold when given $g \in G$, $\sigma \in \Sigma_k$, and $\tau_r \in \Sigma_{j_r}$:

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_{\sigma(1)}, \dots, j_{\sigma(k)}) \circ ((g, \sigma) \times (g, \sigma)^{-1})$$

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_1, \dots, j_k) \circ (\text{id} \times \prod_r (\tau_r \times \tau_r^{-1}))$$

They also need to satisfy the following coherence properties:

(iii) **(Operadic Composition)** The following pasting diagrams are equal, given $j_+ = \sum j_r$, $i_r = \sum_s i_{r,s}$, and $i_+ = \sum_{r,s} i_{r,s}$ with $1 \leq r \leq k$ and $1 \leq s \leq j_r$:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \prod_s (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}})) & \xrightarrow{\text{id} \times \prod (\text{id} \times \theta_{i_{r,s}})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod \theta_{j_r}} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
\downarrow \gamma \times \text{id} & & \downarrow \gamma & & \downarrow \theta_k \\
\mathcal{Q}_G(j_+) \times \prod_{r,s} (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & \xrightarrow{\text{id} \times \prod \theta_{i_{r,s}}} & \mathcal{Q}_G(j_+) \times \mathcal{A}^{j_+} & \xrightarrow{\theta_{j_+}} & \mathcal{A} \\
& \searrow \gamma \times \text{id} & \downarrow \phi & \nearrow \theta_{j_+} & \\
& & \mathcal{Q}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{i_+}} & \mathcal{A}
\end{array}$$

equals

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \prod_s (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}})) & \xrightarrow{\text{id} \times \prod (\text{id} \times \theta_{i_{r,s}})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod \theta_{j_r}} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
\downarrow \gamma \times \text{id} & \searrow \text{id} \times \prod (\gamma \times \text{id}) & \downarrow \text{id} \times \prod \phi & & \downarrow \theta_k \\
\mathcal{Q}_G(j_+) \times \prod_{r,s} (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(i_r) \times \mathcal{A}^{i_r}) & \xrightarrow{\text{id} \times \prod \gamma} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
& \searrow \gamma \times \text{id} & \downarrow \gamma \times \text{id} & \nearrow \phi & \downarrow \theta_k \\
& & \mathcal{Q}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{i_+}} & \mathcal{A}
\end{array}$$

(commuting squares are operadic associativity)

(iv) **(Operadic Identity)** The following two pairs of two pasting diagrams are equal:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \iota^k} & \mathcal{Q}_G(k) \times (\mathcal{Q}_G(1) \times \mathcal{A})^k & \xrightarrow{\text{id} \times \theta_1^k} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
& & \downarrow \text{shuffle} & & \downarrow \theta_k \\
& & \mathcal{Q}_G(k) \times \mathcal{Q}_G(1)^k \times \mathcal{A}^k & & \\
& & \downarrow \gamma \times \text{id} & \nearrow \phi & \\
& & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A}
\end{array}$$

equals

$$\begin{array}{ccccc}
& & \mathcal{Q}_G(k) \times (\mathcal{Q}_G(1) \times \mathcal{A})^k & & \\
& \nearrow \text{id} \times \iota^k & \downarrow \text{id} \times \Pi \phi & \nwarrow \text{id} \times \theta_1^k & \\
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id}} & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A} \\
\downarrow \text{id} \times \iota^k & \searrow \iota^k & \nearrow \gamma \times \text{id} & & \\
\mathcal{Q}_G(k) \times (\mathcal{Q}_G(1) \times \mathcal{A})^k & \xrightarrow{\text{shuffle}} & \mathcal{Q}_G(k) \times \mathcal{Q}_G(1)^k \times \mathcal{A}^k & &
\end{array}$$

(uses operadic associativity)

Also:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \theta_k} & \mathcal{Q}_G(1) \times \mathcal{A} \\
& \searrow \text{id} & \downarrow \gamma \times \text{id} & \nearrow \phi & \downarrow \theta_1 \\
& & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A}
\end{array}$$

equals

$$\begin{array}{ccc}
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{Q}_G(k) \times \mathcal{A}^k \\
\downarrow \theta_k & & \downarrow \text{id} \times \theta_k \\
\mathcal{A} & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{A} \\
& \searrow \phi_1 & \downarrow \theta_1 \\
& & \mathcal{A}
\end{array}$$

(Uses operadic unitality)

Definition 3.7 ($\mathcal{Q}_G$ -pseudomorphism). A $\mathcal{Q}_G$ -pseudomorphism $(\mathbb{F}, \zeta) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ consists of G -functor $\mathbb{F} : \mathcal{A} \rightarrow \mathcal{B}$, such that for each j in the folloigin diagram commutes up to an invertibal natural transformation ζ_j :

$$\begin{array}{ccc}
\mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j \\
\downarrow \theta_j & \nearrow \zeta_j & \downarrow \vartheta_j \\
\mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
\end{array}$$

Moreover, they satisfy the following properties:

- (i) **(Equivariance of ζ)** For each $j, \zeta_j = \zeta_j \circ ((g, \sigma) \times (g, \sigma)^{-1})$ for all $(g, \sigma) \in G \times \Sigma_j$.
- (ii) **(Preserves Identity)** The following two pasting diagrams are equal:

$$\begin{array}{ccc}
\mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} \\
\downarrow \iota & & \downarrow \iota \\
\mathcal{Q}_G(1) \times \mathcal{A} & \xrightarrow{\text{id} \times \mathbb{F}} & \mathcal{Q}_G(1) \times \mathcal{B} \\
& \searrow \theta_1 & \nearrow \zeta_1^{-1} \\
& & \mathcal{A} \\
& & \downarrow \theta_1 \\
& & \mathcal{A} \xrightarrow{\mathbb{F}} \mathcal{B}
\end{array}
=
\begin{array}{ccc}
\mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} \\
\downarrow \iota & & \downarrow \iota \\
\mathcal{Q}_G(1) \times \mathcal{A} & \xrightarrow{\text{id} \times \mathbb{F}} & \mathcal{Q}_G(1) \times \mathcal{B} \\
& \searrow \theta_1 & \nearrow \phi_1 \\
& & \mathcal{A} \\
& & \downarrow \theta_1 \\
& & \mathcal{A} \xrightarrow{\mathbb{F}} \mathcal{B}
\end{array}$$

(iii) **(Preserves Composition)** The following two pasting diagrams are equal:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{B}^{j_r}) & & \\
\downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \swarrow \text{id} \times \prod \vartheta_{j_r} & \searrow \text{id} \times \prod \vartheta_{j_r} & \\
& \mathcal{Q}_G(k) \times \mathcal{A}^k & \xleftarrow{\text{id} \times \prod \zeta_{j_r}} & \mathcal{Q}_G(k) \times \mathcal{B}^k & \\
& \downarrow \theta_k & \xrightarrow{\text{id} \times \mathbb{F}^k} & \downarrow \vartheta_k & \\
\mathcal{Q}_G(j) \times \mathcal{A}^j & \xleftarrow{\phi} & \mathcal{A} & \xrightarrow{\zeta_k} & \mathcal{B} \\
& \searrow \theta_j & \downarrow \theta_j & \swarrow \vartheta_j & \\
& & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
\end{array}$$

equals

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{B}^{j_r}) & & \\
\downarrow \gamma & & \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \\
& \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j & \\
& \searrow \theta_j & \swarrow \zeta_j & \searrow \vartheta_j & \\
& & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
\end{array}$$

Definition 3.8 ($\mathcal{Q}_G$ -2-cells). Let $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ be two $\mathcal{Q}_G$ -pseudomorphisms, a $\mathcal{Q}_G$ -2-cell $\lambda : (\mathbb{F}, \zeta) \Rightarrow (\mathbb{G}, \xi)$ is a natural transformation $\lambda : \mathbb{F} \rightarrow \mathbb{G}$, such that the following two pasting diagrams are equal:

$$\begin{array}{ccc}
\mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j \\
\downarrow \theta_j & \searrow \text{id} \times \lambda^j & \downarrow \vartheta_j \\
& \mathcal{A} & \xrightarrow{\mathbb{G}} & \mathcal{B}
\end{array}
=
\begin{array}{ccc}
\mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j \\
\downarrow \theta_j & \searrow \zeta_j & \downarrow \vartheta_j \\
& \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
\end{array}$$

3.3 2-Category $\mathcal{F}_G\text{-PsAlg}$, Version 1

In the category $\mathcal{F}_G$ (all based finite G -sets), the homsets $\mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ can also be viewed as a discrete category. We'll use this identification to define some notions.

Definition 3.9 ($\mathcal{F}_G$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra, is a pseudofunctor $\tilde{\mathcal{A}} : \mathcal{F}_G \rightarrow \mathbf{Cat}_G$, such that restricting to the full subcategory $\Pi_G \subset \mathcal{F}_G$, $\tilde{\mathcal{A}}$ is a strict functor. Moreover, we require it to be

reduced, namely $\tilde{\mathcal{A}}(0) = *$ (the terminal G -category), and each hom-functor

$$\tilde{\mathcal{A}}_{\mathbf{n}^\alpha, \mathbf{m}^\beta} : \mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) \rightarrow \mathbf{Cat}_G(\tilde{\mathcal{A}}(\mathbf{n}^\alpha), \tilde{\mathcal{A}}(\mathbf{m}^\beta))$$

is a G -functor.

Question: In operad case, we have natural isomorphism between any two G -actions, do we need it here?

Let $\delta_i : \mathbf{n}^\alpha \rightarrow \mathbf{1}$ be the i th projection (namely $i \mapsto 1$, and any $i \neq j \mapsto 0$), since it belongs to Π_G , it generates a functor $\tilde{\mathcal{A}}\delta_i : \tilde{\mathcal{A}}(\mathbf{n}^\alpha) \rightarrow \tilde{\mathcal{A}}(\mathbf{1})$. Hence, by universality of product, all $\tilde{\mathcal{A}}\delta_i$ ($1 \leq i \leq n$) induces a functor $\delta : \tilde{\mathcal{A}}(\mathbf{n}^\alpha) \rightarrow \tilde{\mathcal{A}}(\mathbf{1})^n$.

Definition 3.10 ($\mathcal{F}_{G, \text{vs}}$ and $\mathcal{F}_{G, \mathbb{R}}$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra $\tilde{\mathcal{A}}$ is *very special* (or an $\mathcal{F}_{G, \text{vs}}$ -pseudoalgebra), if all $\delta : \tilde{\mathcal{A}}(\mathbf{n}^\alpha) \rightarrow \tilde{\mathcal{A}}(\mathbf{1})^{\mathbf{n}^\alpha}$ is an isomorphism for all $\mathbf{n}^\alpha \in \mathcal{F}_G$, and $\tilde{\mathcal{A}}$ is *strictly special* (or an $\mathcal{F}_{G, \mathbb{R}}$ -pseudoalgebra), if all δ is in fact identity.

Definition 3.11 ($\mathcal{F}_G$ -1-cell). Given $\tilde{A}, \tilde{B}$ two $\mathcal{F}_G$ -pseudoalgebras, a morphism (or $\mathcal{F}_G$ -1-cell) between them is a pseudo-transformation $(\mathbb{F}, \zeta) : \tilde{\mathcal{A}} \rightarrow \tilde{\mathcal{B}}$, such that restricting to Π_G , it becomes an actual natural transformation.

Fill in details for the commutativity diagrams

Definition 3.12 ($\mathcal{F}_G$ -2-cell). Given $\tilde{A}, \tilde{B}$ two $\mathcal{F}_G$ -pseudoalgebras, and two $\mathcal{F}_G$ -1-cells $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : \tilde{A} \rightarrow \tilde{B}$, an $\mathcal{F}_G$ -2-cell is a transformation $\lambda : \mathbb{F} \Rightarrow \mathbb{G}$, such that for any morphism $f : \mathbf{n}^\alpha \rightarrow \mathbf{m}^\beta$ in $\mathcal{F}_G$, the following two pasting diagrams are equal:

$$\begin{array}{ccc} \begin{array}{ccc} & \xrightarrow{\mathbb{F}_{\mathbf{n}^\alpha}} & \\ \tilde{\mathcal{A}}(\mathbf{n}^\alpha) & \xrightarrow{\lambda} & \tilde{\mathcal{B}}(\mathbf{n}^\alpha) \\ \tilde{\mathcal{A}}f \downarrow & \searrow \mathbb{G}_{\mathbf{n}^\alpha} & \downarrow \tilde{\mathcal{B}}f \\ \tilde{A}(\mathbf{m}^\beta) & \xrightarrow{\mathbb{G}_{\mathbf{m}^\beta}} & \tilde{B}(\mathbf{m}^\beta) \end{array} & = & \begin{array}{ccc} & \xrightarrow{\mathbb{F}_{\mathbf{n}^\alpha}} & \\ \tilde{\mathcal{A}}(\mathbf{n}^\alpha) & \xrightarrow{\lambda} & \tilde{\mathcal{B}}(\mathbf{n}^\alpha) \\ \tilde{\mathcal{A}}f \downarrow & \searrow \mathbb{F}_{\mathbf{m}^\beta} & \downarrow \tilde{\mathcal{B}}f \\ \tilde{A}(\mathbf{m}^\beta) & \xrightarrow{\lambda} & \tilde{B}(\mathbf{m}^\beta) \end{array} \end{array}$$

4 Construction of $\mathbb{R}_G$

5 Counterexample: $\mathcal{F}_{G,\mathbb{R}}$ -pseudoalgebra that's not $\mathcal{Q}_G$ -pseudoalgebra

Here we fix the group $G := \mathbb{Z}/2\mathbb{Z}$. Which, we have $\mathcal{Q}_G(1) = \mathcal{P}_G(1) = *$, in particular $G \times \Sigma_1 \cong G$ has trivial action on these. This leads to the following properties of $\mathcal{Q}_G$ or $\mathcal{P}_G$ -pseudoalgebra:

Proposition 5.1. *For any $\mathcal{Q}_G$ -pseudoalgebra $\mathcal{A}$, any G -action $g : \mathcal{A} \xrightarrow{\sim} \mathcal{A}$ is naturally isomorphic to the identity functor $\text{Id}_{\mathcal{A}}$.*

Proof. Pasting the unitality and equivariance diagram together, we get the following naturality diagram:

$$\begin{array}{ccc}
 \mathcal{A} & \xrightarrow{g^{-1}} & \mathcal{A} \\
 \downarrow \iota & & \downarrow \iota \\
 \mathcal{Q}_G(1) \times \mathcal{A} & \xrightarrow{g \times g^{-1}} & \mathcal{Q}_G(1) \times \mathcal{A} \\
 \downarrow \theta_1 & & \downarrow \theta_1 \\
 \mathcal{A} & \xrightarrow{\text{Id}_{\mathcal{A}}} & \mathcal{A}
 \end{array}
 \quad (5.2)$$

Here, the top trapezoid commutes because G acts trivially on $\mathcal{Q}_G(1)$. This identifies a natural isomorphism as follow:

$$g^{-1} = \text{Id}_{\mathcal{A}} \circ g^{-1} \xrightarrow{\phi_1^{-1} * \text{id}_{g^{-1}}} (\theta_1 \circ \iota) \circ g^{-1} = \theta_1 \circ (g \times g^{-1}) \circ \iota = \theta_1 \circ \iota \xrightarrow{\phi_1} \text{Id}_{\mathcal{A}} \quad (5.3)$$

□

To construct a counterexample for our fixed G , it suffices to find an $\mathcal{F}_{G,\mathbb{R}}$ -pseudoalgebra that doesn't have the above property. Here, we largely use the fact that for two naturally isomorphic functors, if the target category is discrete, the two functors must be equal.

Construction 5.4. Let $\mathcal{A} := \{0, 1\}$ denotes the discrete category with 2 objects, which a G -action on $\mathcal{A}$ is specified by its action on the objects due to its discreteness. Define a G -action on $\mathcal{A}$ given by

$$H_0 : \mathcal{A} \xrightarrow{\sim} \mathcal{A}, \quad H_0(0) = 0, \quad H_0(1) = 1 \quad (5.5)$$

$$H_1 : \mathcal{A} \xrightarrow{\sim} \mathcal{A}, \quad H_1(0) = 1, \quad H_1(1) = 0, \quad (5.6)$$

Which, $H_0^2 = H_0$, $H_0 \circ H_1 = H_1 \circ H_0$, and $H_1^2 = H_0$, so H_0, H_1 satisfies the relations of $G = \mathbb{Z}/2\mathbb{Z}$, hence inducing a G -action on $\mathcal{A}$.

Define $\tilde{\mathcal{A}} : \mathcal{F}_G \rightarrow \mathbf{Cat}_G$ given by $\tilde{\mathcal{A}}(\mathbf{n}^\alpha) := \mathcal{A}^{\mathbf{n}^\alpha}$ (with underlying category $\mathcal{A}^n$, a discrete category with 2^n objects), and for any morphism $f : \mathbf{n}^\alpha \rightarrow \mathbf{m}^\beta$, define $\tilde{\mathcal{A}}f : \mathcal{A}^{\mathbf{n}^\alpha} \rightarrow \mathcal{A}^{\mathbf{m}^\beta}$ by

$$\tilde{\mathcal{A}}f((a_i)_{i=1,\dots,n}) := \left(\sum_{i \in f^{-1}(j)} a_i \bmod 2 \right)_{j=1,\dots,m}, \quad (5.7)$$

for special case where $f^{-1}(j) = \emptyset$, define the j th entry as 0.

To check it preserves composition up to coherent natural isomorphism, consider $\mathbf{n}^\alpha \xrightarrow{f} \mathbf{m}^\beta \xrightarrow{g} \mathbf{k}^\gamma$ in $\mathcal{F}_G$, then $\tilde{\mathcal{A}}(g \circ f) : \tilde{\mathcal{A}}^{\mathbf{n}^\alpha} \rightarrow \tilde{\mathcal{A}}^{\mathbf{m}^\beta}$ is given as follow:

$$\tilde{\mathcal{A}}(g \circ f)((a_i)_{i=1,\dots,n}) = \left(\sum_{i \in (g \circ f)^{-1}(\ell)} a_i \bmod 2 \right)_{\ell=1,\dots,k} \quad (5.8)$$

$$= \left(\sum_{i \in f^{-1}(g^{-1}(\ell))} a_i \bmod 2 \right)_{\ell=1,\dots,k} \quad (5.9)$$

$$= \left(\sum_{j \in g^{-1}(\ell)} \left(\sum_{i \in f^{-1}(j)} a_i \bmod 2 \right) \bmod 2 \right)_{\ell=1,\dots,k} \quad (5.10)$$

$$= \tilde{\mathcal{A}}g \left(\left(\sum_{i \in f^{-1}(j)} a_i \bmod 2 \right)_{j=1,\dots,m} \right) \quad (5.11)$$

$$= \tilde{\mathcal{A}}g \left(\tilde{\mathcal{A}}f((a_i)_{i=1,\dots,n}) \right) \quad (5.12)$$

(Here, the well-defineness of the function heavily relies on the assumption that if $f^{-1}(j) = \emptyset$, the corresponding sum is 0). In particular, $\tilde{\mathcal{A}}(g \circ f) = \tilde{\mathcal{A}}g \circ \tilde{\mathcal{A}}f$, so $\tilde{\mathcal{A}}$ is a functor (or an $\mathcal{F}_G$ -algebra), and since each $\tilde{\mathcal{A}}(\mathbf{n}^\alpha) := \tilde{\mathcal{A}}^{\mathbf{n}^\alpha}$, it's in fact an $\mathcal{F}_{G,\mathbb{R}}$ -algebra.

Yet, since $\mathcal{A}$ is discrete, with H_0, H_1 being distinct functors on $\mathcal{A}$, we see that not all the G -actions on $\mathcal{A}$ are naturally isomorphic to $\text{Id}_{\mathcal{A}}$, violating the property described in previous proposition. Hence, $\mathcal{A}$ cannot be a $\mathcal{Q}_G$ -pseudoalgebra, or even a $\mathcal{P}_G$ -pseudoalgebra due to the same reasoning.

This shows that the functor $\mathbb{R}_G : \mathcal{P}_G\text{-PsAlg} \rightarrow \mathcal{F}_G\text{-PsAlg}$ is not essentially surjective, hence cannot be an equivalence of categories. Same statement holds true for $\mathcal{Q}_G\text{-PsAlg}$.

5.1 Extra Axioms Required on $\mathcal{F}_G\text{-PsAlg}$

6 Construction of $\mathbb{Q}_G$ under Extra Axiom

7 (If everything works out) $\mathbb{R}_G$ and $\mathbb{Q}_G$ are Isomorphisms

References

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